INNOVATION DESIGN PROJECT • ZEROTH REVIEW

CloudGuardian AI

An AI-Based Self-Healing Cloud Infrastructure for Predictive Failure Detection and Automated Service Recovery

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Problem Statement & Proposed Solution

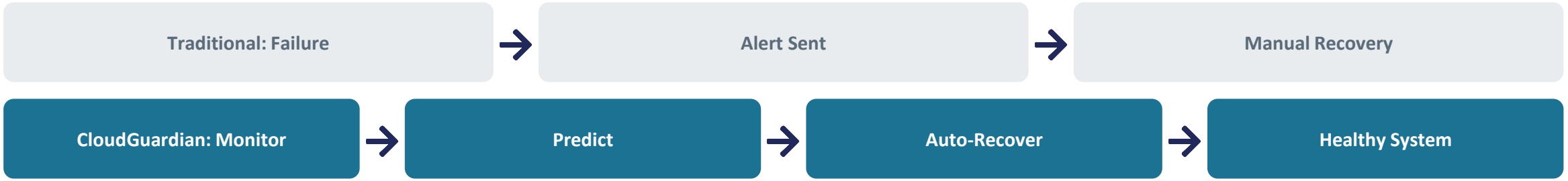
Current Problems

- Downtime after failures occur
- Manual recovery required
- Slow, reactive detection
- Increased operational costs
- Reduced service availability

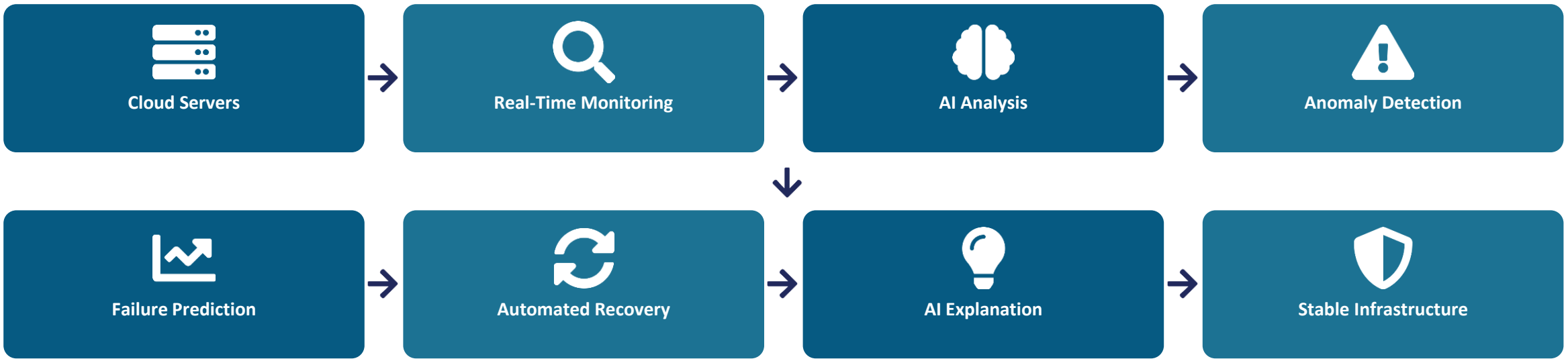
Our Proposed Solution

- Continuous AI-based monitoring
- Predictive failure detection
- Automated recovery actions
- Intelligent decision making
- Minimized downtime

Traditional vs. CloudGuardian AI Approach



Project Overview & Working



Key Project Objectives

- Predict Failures Early
- AI-Driven Anomaly Detection
- Reduce Downtime
- Automate Recovery
- Improve Availability
- Minimize Human Effort

Technology Stack

Planned tools and frameworks across the system

Cloud Computing

- AWS (Basics)

Artificial Intelligence

- Machine Learning
- Anomaly Detection

DevOps

- Docker
- Kubernetes (Basics)

Backend

- Node.js
- Express.js

Database

- MongoDB

Frontend

- React.js
- Tailwind CSS

Software Engineering

- System Design

Project Development Roadmap



PHASE 1

Research

- Study cloud computing basics
- Review research papers & patents

PHASE 2

Problem ID

- Identify system limitations
- Finalize novelty & scope

PHASE 3

System Design

- Cloud & AI architecture
- API and database planning

PHASE 4

Prototype

- Build monitoring & AI modules
- Develop recovery + dashboard

PHASE 5

Testing & Patent

- Performance evaluation
- Patent writing & documentation

This roadmap represents the planned development lifecycle of our proposed intelligent cloud infrastructure.

THANK YOU